

A Tutorial in Dynamic Systems Simulation and Modelling With R, ctsem, and lme4: Couples' Affect Dynamics Over Time

To editor

This manuscript presents a tutorial on dynamic systems simulation and modeling using R, ctsem, and lme4, organized around a running example of couples' affect dynamics over time. It is ambitious in scope, pedagogically oriented, and clearly intended to help readers move from familiar mixed-effects models toward more complex dynamic systems formulations. The emphasis on simulation, visualization, and iterative model-building is potentially valuable. My central concern is that the paper remains substantially closer to an extended software tutorial than to a strong methodological article. It demonstrates many modeling possibilities, but it does not yet distill them into a sufficiently clear, general, and high-level methodological contribution. Overall, I have some major concerns about this paper.

Main concerns:

1. The manuscript does not yet establish a sufficiently strong methodological contribution beyond package documentation. At present, the paper reads primarily as a long implementation-oriented tutorial. It walks the reader through code, model setup, and progressively more complex examples, but it does not clearly demonstrate what the paper contributes methodologically beyond what could be learned from package vignettes, workshop materials, or a well-developed software tutorial. The authors need to extract and explicitly articulate a small set of transferable principles, rather than relying on the accumulation of examples to imply the contribution.
2. The paper's generalizability is limited by its heavy reliance on a single didactic example. The running example of couples' affect dynamics is pedagogically effective, but the manuscript remains too tightly tied to this particular scenario. As a result, the paper sometimes feels less like a broad methodological tutorial and more like a long, well-developed demonstration within one narrow substantive context.
3. The tutorial is built largely around hypothetical or simulated data-generating mechanisms. This is understandable for pedagogical purposes, but it also limits the manuscript's practical credibility. Real psychological data often involve irregular spacing, substantial missingness, nonstationarity, limited measurement precision, floor/ceiling effects, non-Gaussian variables, and model misspecification that is more ambiguous than the examples shown here. The manuscript needs a dedicated

section on scope conditions, empirical limitations, and practical failure points. At minimum, the authors should discuss: when these models are likely to be appropriate, when simpler models may be preferable, what kinds of data are insufficient for the more complex models, and what common empirical problems are not addressed by the tutorial.

4. The manuscript repeatedly frames the approach as continuous-time SEM and state-space modeling. However, much of the tutorial relies on highly simplified single-indicator setups. That may be acceptable pedagogically, but the paper does not adequately acknowledge the consequences of that simplification. As written, the paper often reads more like a tutorial on continuous-time dynamic regression with measurement error than a tutorial that fully leverages the latent-variable advantages of continuous-time SEM.
5. The manuscript needs stronger guidance on assumptions, identification, and model choice. The tutorial is effective at demonstrating how one can move from simple to more complex models. However, it is far less effective at telling readers when they should do so, and on what basis. This is a serious omission. A strong methods tutorial should not only demonstrate that a model can be fit; it should also help the reader decide: when linear growth is inadequate, when state dependence is warranted, when system noise should be distinguished from measurement error, when interactions or time-varying moderation are theoretically justified, and when data are too weak to support these expansions.
6. The later sections on misspecification and posterior predictive checks are among the better parts of the manuscript. However, the paper still does not provide enough practical guidance on the difficulties of estimation. The authors should add a practical troubleshooting section that explicitly discusses common failure modes and how readers should respond to them. The paper should do more to teach caution, not only implementation.
7. The authors note in the cover letter that the manuscript is longer than typical submissions because of figures and code. I agree that the paper is currently overlong for its central contribution. The issue is not merely page count; it is also information allocation. Too much of the main text is occupied by details that could be moved to supplementary material without reducing the paper's conceptual value.
8. The conclusion is notably underdeveloped. It does not adequately distill what the reader should have learned, nor does it clearly summarize the limits of the proposed workflow.

I also have a list of specific comments provided to the authors.

Specific comments are given below:

1. There are multiple language and copyediting problems that should be corrected before the paper can be seriously evaluated in final form. A few examples include: “introduce a approaches”, “vector autogression”, “do not be mislead”, “relatively poor”.
2. Some code examples appear inconsistent or insufficiently cleaned. Because this is a tutorial, code accuracy and consistency matter especially strongly. I noticed at least one example in which parameter assignments appear contradictory (e.g., `B <- .5` followed immediately by `B <- 2` in the same continuous-time simulation block). Even if harmless in context, such issues undermine reader trust in the tutorial.